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CSC 4305 · Software Engineering · Fall 2026

Plan-Driven Process Models I — Waterfall and the V-model

Two models that sign things off before moving on, and what the choice costs when it is wrong

Session 3 of 28 · Tuesday 15 September 2026 · 1:00 – 2:20 PM

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Plan-Driven Process Models I — Waterfall and the V-model

SESSION 3 OF 28

Tuesday, 15 September 2026 · 1:00 – 2:20 PM · Unit 1 · Foundations and Process

LEARNING OBJECTIVES

- ✓ Define waterfall, name its phases in order, and state the four assumptions that make it a rational choice.
- ✓ Walk SANTE-BASSAM through each waterfall phase and name the artifact each phase must sign off.
- ✓ Explain the V-model as specification levels paired with test levels.
- ✓ Write a test at each of the four levels, tracing to a named SANTE-BASSAM requirement.
- ✓ Read a traceability matrix and find the gap in it.

FROM SESSION 2

You made one architectural bet — offline-first, dated 11 September — and wrote down three numbers you can be judged on in December. Both of those are predictions. Today asks the question every process model is an answer to: **when is the team allowed to find out it was wrong?**

Baseline the numbers, and record the decision

Applying Session 2. Q2 and Q4 were answered on Friday. Today they become artifacts with dates on them, because an undated answer cannot be checked in December.

Pair A — the baseline

write `plan/baseline.md`: the three Q2 measures, today's value, the December target, and **exactly how each will be measured**, including who holds the stopwatch

Pair B — the decision record

write `arch/ADR-001-offline-first.md`: context, the two options, the choice, the cost-if-wrong for each, the date, and who agreed

Pair C — the visit

fix the date of the first visit to the post, what you will do there, and what you will bring back. One paragraph, in the log

1. **Six minutes** in pairs. **Six minutes** reporting — Pair C last, because the visit date constrains the other two.
2. **Three minutes**: agree one sentence that would tell you, in December, that ADR-001 was the wrong call. If you cannot write it, you have not made a decision — you have made an assumption.

Output: two files committed, one visit date in the log, and the falsifying sentence.

Next session's slot: the traceability matrix you build today, extended and owned.

1 · The one question every process model answers

Five process models, five different answers, and one way to compare them. Take a single fact about SANTE-BASSAM that the team does not yet know — *the clerk searches by the mother's name* — and ask, for each model, in which week the team finds out.

“The clerk searches by the mother’s name.” — in which week does the team find out?

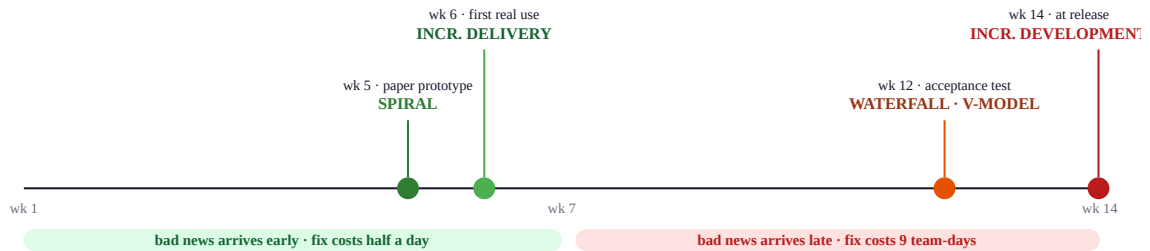


Figure 1. Same team, same project, same surprise — only the week it arrives changes. A process model is a decision about when the team is allowed to find out it was wrong; everything else in these two sessions is the mechanics of moving that moment to the left.

TWO FACTS ABOUT SANTE-BASSAM, AND THEY PULL IN OPPOSITE DIRECTIONS

Fixed by law: the identifier format, consent for minors, record retention. These *cannot* be discovered by iterating — you read the regulation once and comply.

Unknowable on paper: the front-desk workflow. Nobody, including the clerk, can describe it precisely until they see software.

Any process you choose has to accommodate both at once. Every model in these two sessions handles one of them well and the other badly, which is why the answer at the end of Friday is not a model but a **line**.

2 · The waterfall model

ONE RULE

Each phase finishes with a **signed artifact** before the next begins. Going back is allowed, but treated as a defect rather than as the normal way of working. (Royce, 1970 — named for the shape of the diagram, and by an author who was describing its weaknesses.)

Phase	Weeks	Exit criterion — what must be true and signed	Signed by
1 · Requirements	1–3	SRS, numbered UR and SR	nurse-in-charge

2 · Design	4–5	Architecture and data model	team lead + external reviewer
3 · Implementation	6–10	Code and unit tests, reviewed	team lead
4 · Integration and test	11–12	Acceptance-test report	nurse-in-charge
5 · Deployment	13	The clerk registers ten real patients	nurse-in-charge
6 · Maintenance	14 →	Numbered versions, change log	system owner

Going back means re-opening and re-signing every artifact in between. That is what makes late change expensive, and it is deliberate — the cost is the mechanism.

2.1 What waterfall assumes

Requirements can be known up front

...and stabilised before design begins.

Late change is expensive

...so front-loading the analysis is rational.

The customer can judge a document

...and tell you whether the specification is right.

Hand-offs beat conversation

...because the team is large or spread across organisations.

When the assumptions hold

Regulated or safety-critical work; contract development where scope defines payment; systems integration with interfaces fixed a year ahead. Waterfall is a **sound choice**, and pretending otherwise is fashion, not judgement.

When they fail

Whenever the customer will only recognise what they want by seeing it. Waterfall then discovers the wrong requirement at the most expensive possible moment: **acceptance testing**.

APPLIED TO SANTE-BASSAM

Fourteen weeks of pure waterfall — and the week-12 discovery

Week 12, acceptance test. A mother arrives without the card. The clerk says: “*I look them up by the mother’s name and village.*” Nobody wrote that down in week 2, because nobody asked. One afternoon at the desk in week 3 would have produced the same fact.

Found in week 3 — watching the clerk

0.5 team-days

Found in week 12 — at the acceptance test

9 team-days

re-sign the SRS · data model + migration · code in three modules ·
four levels of new tests · re-run the whole acceptance campaign

18× more expensive — in the two weeks you had left

What waterfall buys

An auditable trail. If the Ministry asks which test proves a power-cut record is not lost, the SRS names UR-2, the report names AT-02, and the nurse’s signature is on both. A health-record system is obliged to produce exactly this evidence.

What waterfall bets

That week-3 elicitation is good enough that week-12 surprises are rare. **Safe for the parts fixed by law. Not safe for the front-desk workflow.**

CARRY FORWARD

Waterfall converts uncertainty about requirements into **one large risk that lands at the end**. Every other model in these two sessions is a different way of moving that risk earlier.

3 · The V-model: specification and test as mirror images

ONE RULE

Every level of specification on the left arm is verified by exactly one level of test on the right — and **the test is written when the level is specified**, not when the code exists.

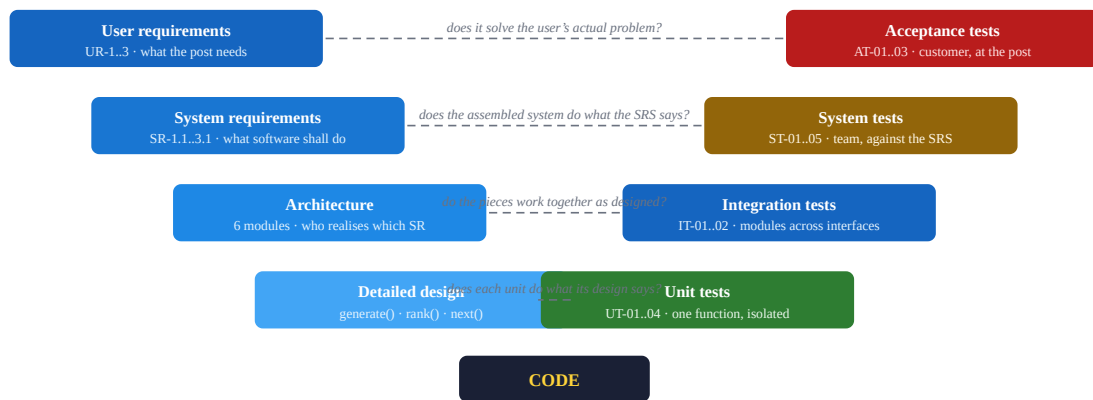


Figure 2. The payoff is **bidirectional traceability**. For any test, which requirement? For any requirement, which test? A requirement with no test is *unverified*; a test with no requirement is either *waste* or a *hidden requirement*. Both are defects, and the *V* makes them visible — which is why it dominates medical-device and avionics work.

APPLIED TO SANTE-BASSAM

Three user requirements, refined down and tested back up

UR-1

The clerk can find an existing patient's record in under one minute, even with no card.

UR-2

A record captured during a power cut or offline is never lost.

UR-3

The district officer gets the monthly report without anyone re-counting the paper register.

SR	Refined system requirement
SR-1.1	Find patients by card number, by family and given name, or by mother's name and village.
SR-1.2	Name search tolerates spelling variants; candidates ranked, with village and birth year shown.
SR-1.3	First page of a 10,000-record search within 2 s on the target laptop.
SR-2.1	Every registration and visit persisted locally before the confirmation is shown.
SR-2.2	Offline records marked pending, uploaded in creation order, synced only after server acknowledgement.
SR-2.3	A record failing upload three times is flagged for supervisor review and kept on the device.
SR-3.1	Monthly report of visits by outcome — attended, missed, rescheduled — per village.

ONE TEST AT EACH LEVEL — THE OTHER TEN FOLLOW THE SAME SHAPE

AT-01 · ACCEPTANCE — FIND A RETURNING PATIENT WHO HAS NO CARD · VERIFIES UR-1

GIVEN the register is migrated; a mother arrives with her child but no card
WHEN the clerk asks only what she would normally ask
(child's name, mother's name, village)
THEN the correct record opens within 60 s in at least 9 of 10 trials
with real returning patients

ST-03 · SYSTEM — OFFLINE CAPTURE, ORDERED SYNC, ACKNOWLEDGEMENT · VERIFIES SR-2.1, SR-2.2

GIVEN the tablet in airplane mode
WHEN three patients are registered, then connectivity returns for 30 s
THEN each shows pending on save; the server receives them in creation order; each becomes synced only after the acknowledgement is logged

IT-02 · INTEGRATION — SYNCQUEUE AGAINST SERVERAPI · VERIFIES THE SYNC CONTRACT

GIVEN 5 pending records; ServerAPI running locally
WHEN sync is triggered
THEN 5 POSTs arrive in queue order; each acknowledgement carries the record id; SyncQueue marks them synced in the same order

UT-03 · UNIT — SYNCQUEUE.NEXT() ORDERING AND GIVE-UP RULE · VERIFIES DETAILED DESIGN

GIVEN A, B, C enqueued in order; A fails every attempt
WHEN next() is called repeatedly and failures are recorded
THEN A is returned 3 times, then B; A's status is needs-review; C follows B

The common student error

Writing all four levels as variations of one scenario. Look down the UR-2 column: **AT-02** pulls a plug in front of the nurse; **ST-03** scripts airplane mode; **IT-02** checks the order of HTTP calls; **UT-03** checks a queue rule with no input or output at all. Each level answers a different question and catches a different defect. If two of your tests would fail for the same reason, one of them is not doing its job.

3.1 The traceability matrix — the artifact the V exists to produce

Requirement	AT	ST	IT	UT	Status
UR-1 → SR-1.1, 1.2, 1.3	AT-01	ST-01, 02	IT-01	UT-02	covered
UR-2 → SR-2.1, 2.2, 2.3	AT-02	ST-03, 04	IT-02	UT-01, 03	covered
UR-3 → SR-3.1	AT-03	ST-05	— GAP —	UT-04	no IT

READING THE LAST ROW IS THE WHOLE SKILL

UR-3 has no integration test, which means **nobody has checked that ReportGenerator reads the same Visit records SyncQueue wrote**. Found in week 2 by looking at a table: five minutes. Found in week 13: a wrong report on the district officer's desk, and a nurse who stops trusting the system.

SESSION 3 TAKEAWAY

Waterfall converts uncertainty about requirements into one large risk that lands at the end, and buys in exchange an auditable trail that a Ministry can read. The V-model adds one discipline — write the test when you write the specification — and thereby moves one particular kind of bad news to week 2: the hole in the test plan. Neither moves the requirements risk itself. Friday is about three models that do.

4 · In-class exercise · close the gap

Twelve minutes in three pairs, four minutes reporting. The matrix above has a hole in it.

Three tasks, one per pair

Pair A

write

IT-03

— ReportGenerator reading the Visit records SyncQueue wrote. Given / When / Then, in the shape of the four worked tests. Name the two modules it exercises together

Pair B

add

UR-4

, appointments. Refine it into two system requirements and name the module that realises each

Pair C

write

one test at each of the four levels

for UR-4 — each answering a different question. Re-read the warning box before you start

Then, together: each pair defends one test — which of yours would catch a defect that none of the other three could? Two minutes each.

EXPECT THIS

Most groups write the acceptance test and the system test as the same scenario at two speeds. The integration test is the one that forces you to name an interface; the unit test is the one with no input or output at all. If you cannot tell your AT from your ST, look at who runs it and where.

5 · Homework · individual · due before Session 4 (Friday 18 September)

The first individual assignment of the course. Yours alone, about forty-five minutes, and nothing in it is hard — it continues the exercise above.

TEMPLATE

Download **CSC4305_HW03_Template.docx** from **CSC_4305_SWE-Assignments**, fill it in, **rename** it **CSC4305_HW03_FirstName_LastName.docx** and **commit** it to **/homework/S03/** in your own private repository.

A .docx file. Not a PDF, not a link. Full instructions are in the *Read this first* box at the top of the template.

Q1 Write the four tests your pair did not write. Whichever of IT-03, the UR-4 refinement, or the four-level set your pair did not take in class — do it alone now.

Given / When / Then throughout. Name the requirement each one verifies.

Q2 Tell the levels apart. Here are four statements about SANTE-BASSAM. For each, say which test level it belongs to and why no other level would catch it.

(a) The search box sends the village as an empty string when nothing is chosen. (b) The clerk cannot tell which of two results is the right child. (c) The report totals are right but the villages are in the wrong order. (d) rank() returns 91 for a one-letter misspelling and 89 for an exact match.

Q3 Find your own gap. Draw the traceability matrix for UR-1 to UR-4 with every test now written — yours and the class's. Mark any remaining cell that is empty, and say in one sentence what defect could reach production through it.

Q4 Argue for waterfall. Three or four sentences, seriously. Name one part of SANTE-BASSAM where pure waterfall is the *right* choice, and say what you would lose by running that part in sprints instead.

There is a good answer [here](#). A submission that says waterfall is simply outdated has not understood §2.1.

Q5 One prediction. Look at ADR-001, the offline-first decision your team recorded this morning. In two sentences: what is the earliest week you could discover it was wrong, under waterfall — and what would you have to do differently to find out sooner?

Friday answers this properly. Write your guess before you hear it.

Q6 Reading. Sommerville, *Software Engineering* 10th ed., §2.1 (waterfall, incremental) and §2.2 (process activities). Come to Friday with **incremental delivery defined in your own words** — Session 4 leaves that one for you.

Key terms

waterfall model

phase sign-off

exit criterion

V-model

acceptance test

system test

integration test

unit test

bidirectional traceability

traceability matrix

architecture decision record

Readings and references

- Sommerville, I. *Software Engineering*, 10th ed. — Chapter 2, §2.1–2.3.

- Royce, W. “Managing the Development of Large Software Systems”, Proc. IEEE WESCON, 1970.
- Forsberg, K. & Mooz, H. “The Relationship of System Engineering to the Project Cycle”, NCOSE, 1991 — the V-model.
- Pressman, R. & Maxim, B. — Chapter 2.

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Next · Session 4 — Plan-Driven Process Models II: Increments, the Spiral and Choosing